

Brain-to-Vision Reconstruction

RelaxedAsCoal — AML Project Report 2025/26

Niccolò Siciliano
Matricola: 1958541

Luiz Eduardo Leite Filho
Matricola: 2219191

Gabriele Gimelli
Matricola: 1950107

Gabriele Moretti
Matricola: 1958932

Abstract

Reconstructing visual stimuli from fMRI data is a complex task due to the high dimensionality and inherent noise of neural signals. In this work, we aimed to replicate the MindEye architecture using the Natural Scenes Dataset (NSD). While we initially intended to implement the full reconstruction pipeline, computational restrictions required us to focus our implementation on the mapping of 15,724 fMRI voxels into the 1,280-dimensional Kandinsky-CLIP space using a simpler funnel-shaped MLP Encoder. The model successfully learns the semantic geometry of the latent space, outperforming baseline regression methods. Our results demonstrate moderate categorical fidelity in the reconstructed images. However, performance remains below the benchmarks set by the original MindEye study. Furthermore, we observed significant spatial inaccuracies, which we attribute to the "pixel-agnostic" nature of CLIP embeddings. Although the original MindEye framework addresses this by extracting low-level structural information from voxels via a separate Variational Autoencoder (VAE) pipeline, this component was omitted in our simplified version due to resource constraints.

1 Introduction

Reconstructing images from brain activity represents one of the most ambitious challenges at the intersection of neuroscience, computer vision, and machine learning. With the advent of large-scale neuroimaging datasets such as the Natural Scenes Dataset (NSD) [1], research has evolved from simple classification to complex reconstruction of natural images from fMRI signals.

State-of-the-art methodologies have shown that achieving high-fidelity reconstruction requires a synergy between semantic meaning and structural detail. Current approaches, such as the MindEye framework, adopt a dual-pipeline strategy that bifurcates the decoding process. One pipeline projects brain signals into a shared latent semantic space, specifically utilizing CLIP embeddings [2] to capture high-level categorical information, a second pipeline focuses on extracting low-level visual features directly from the voxels to guide the spatial arrange-

ment and geometric structure of the image, often through the use of Variational Autoencoders (VAE).

In this project, we implemented a funnel-shaped MLP Encoder designed to progressively distill information from the 15,724 NSD voxels down to the 1,280 dimensions of the Kandinsky-CLIP space [3]. The main innovation lies in the training architecture: we did not limit ourselves to minimizing pointwise prediction error, but employed a combined loss that integrates contrastive, triplet, and cosine regression components. This approach allows the model to learn the true geometry of the semantic space, aligning fMRI signals with the correct visual concepts.

The work documents how such alignment allows preserving the identity of subjects in reconstructed images, while also discussing the importance of integrating low-level structural features to improve the spatial consistency of reconstructions.

2 Related work

Brain decoding has progressed from linear classification to deep learning-based reconstruction. While mapping neural activity to CNN features improved performance, direct voxel-to-pixel regression remains limited by the high dimensionality and low signal-to-noise ratio of fMRI data, often resulting in blurry output.

A key breakthrough involves utilizing semantic representations like CLIP embeddings. By aligning fMRI signals with a shared vision-language latent space, these methods improve decoding coherence over pixel reconstruction. The MindEye framework [4] established this via a two-stage pipeline: projecting fMRI to CLIP space, then generating images with latent diffusion models. MindEye2 [5] recently standardized this paradigm with rigorous benchmarks for training strategies.

Metric Learning techniques, like contrastive and triplet losses, are ideal for cross-modal alignment by enforcing distance metrics where semantically related samples cluster together. Our project adopts this framework, optimizing a funnel-shaped MLP Encoder to capture the semantic structure of Kandinsky's embedding space. This strategy leverages pre-trained generative models to bypass the hardware limitations of end-to-end reconstruction.

3 Proposed method

The proposed method adopts the brain-to-vision paradigm introduced by MindEye, focusing on the effectiveness of semantic mapping between fMRI signals and vision-language representations. The pipeline consists of three stages: signal pre-processing, fMRI-to-CLIP encoding, and image reconstruction using diffusion models (in this case using pre-trained models).

3.1 fMRI-to-CLIP Semantic Mapping

Given an fMRI activation vector $x \in \mathbb{R}^{15724}$ of the visual cortex, the goal is to predict a semantic embedding $\hat{z} \in \mathbb{R}^{1280}$ in the latent CLIP space compatible with the Kandinsky model. The architecture consists of a funnel-shaped MLP encoder that operates a progressive distillation of the signal through five hierarchical layers with decreasing dimensions [12000, 10000, 8192, 4096, 2048]. This stepwise compression is designed to filter out the high intrinsic noise of voxels, acting as a distillation mechanism that preserves relevant semantic information. To prevent overfitting and stabilize training, we have integrated 20% dropout and weight decay.

3.2 Loss Function and Training Objective

Unlike standard linear regression, we adopt a metric learning approach to capture the complex geometry of the CLIP space. The total loss is a weighted combination of three terms:

- Contrastive Loss: optimizes the alignment between fMRI-image pairs within the batch.
- Triplet Loss: enforces a margin of separation between correct and incorrect matches.
- Cosine Regression: maximizes the directional similarity between predicted and target embeddings.

This synergy allows the model to learn the relational structure of the semantic space, prioritizing categorical coherence over mere pointwise numerical accuracy.

3.3 Image Reconstruction

The decoding phase uses the pre-trained Kandinsky diffusion model as a fixed decoder. We chose not to fine-tune the diffusion model in order to purely isolate and analyze the effectiveness of the semantic mapping. Although MindEye2 integrates low-level spatial conditioning, our approach focuses on the CLIP-based pathway to assess how much deep conceptual information can be extracted directly from the neural signal, leaving the integration of structural features for future development.

4 Dataset and Benchmark

4.1 Dataset

The dataset used contains fMRI recordings from human subjects while they were viewing natural images from the COCO dataset [4]. Each image is shown three times to the same subject. Each viewing instance, called a trial, is represented as a high-dimensional vector of brain activity, with around fifteen thousand voxels. These repeated measurements capture the intrinsic noise and variability of fMRI signals.

To avoid data leakage, the dataset is split at the image level rather than at the trial level. This means that all fMRI repetitions of a given image appear either in the training set or in the validation set, but never in both.

During training, the model operates at the trial level: each individual fMRI trial is mapped to the CLIP embedding of the image that was shown. Treating repetitions as independent samples increases the effective amount of training data (8 000 unique images, 24 000 fMRI trials) and helps the model learn a more robust mapping from brain activity to visual semantics.

At validation time, predictions are aggregated at the image level. We average the predicted embeddings across all trials of the same image, which reduces noise and provides a more stable representation for evaluation (2 000 unique images, 6 000 fMRI trials).

4.2 Benchmark Task

The main task is brain semantic decoding: given an fMRI activation x , the model must predict the corresponding CLIP $\hat{z} \in \mathbb{R}^{1280}$ embedding. Instead of focusing on pixel reconstruction error, performance is evaluated in latent space via similarity metrics and retrieval targets. This methodological choice reflects the nature of the fMRI signal, which is more suitable for decoding semantic concepts than for reconstructing low-level visual details. Finally, these embeddings act as a conditioning input for the diffusion decoder in the image reconstruction phase.

5 Experimental results

We evaluate our approach through both quantitative and qualitative analyses, in order to assess not only numerical performance in the latent space, but also the semantic and visual quality of the reconstructed images.

5.1 Quantitative Analysis

The performance is measured in the CLIP embedding space through cosine similarity metrics (comparing the true embedding generated from the original image and the embedding generated from the fMRI signal).

The results show that the combination of the MLP encoder and metric learning losses consistently outperforms a standard regression baseline. The model reaches stable convergence, effectively capturing the semantic structure of the target.

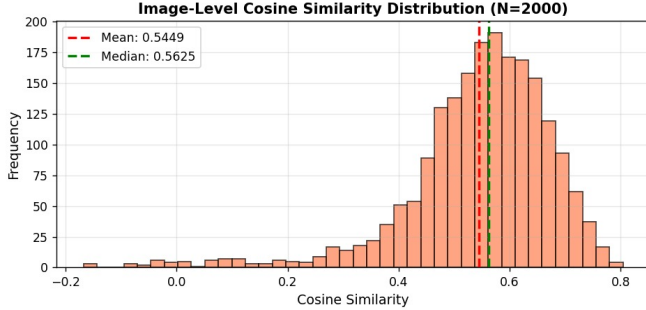


Figure 1: Quantitative Results of Trained Translation Model

5.2 Qualitative Analysis

The qualitative evaluation, performed using the Kandinsky diffusion model conditioned on the predicted embeddings, demonstrates that the reconstructions generally preserve high-level semantic content such as general object categories and scene types. This indicates that the MLP encoder successfully learned the mapping from fMRI signals to the CLIP latent space.

However, notable inaccuracies persist in the semantic and spatial structure of the reconstructions. In some instances, the model captures the core components of an action but fails to accurately identify the specific environment. For example, a target image of a person on a beach was reconstructed as a person skiing, which suggests that while the model recognizes a person in an open landscape, the specific environmental cues are sometimes conflated within the latent space. Furthermore, significant inaccuracies are observed in fine grained spatial details and object localization, particularly in crowded scenes. These reconstructions often appear semantically plausible but spatially ambiguous, confirming that relying solely on CLIP embeddings overlooks critical low level structural constraints.

In our implementation, we attempted to mitigate these low level discrepancies by using a blurred version of the original image as a structural anchor to provide a starting point for the diffusion process. This differs from the original MindEye framework, which employs a dedicated VAE pipeline to extract structural information directly from the fMRI voxels. Without this direct neural decoding of low level features, our reconstructions rely more heavily on the generative prior of the diffusion model. Additionally, we observed a consistent blue shift in the outputs, a known characteristic of the Kandinsky diffusor that can introduce chromatic inaccuracies, further distancing the reconstructions from the original visual stimuli.

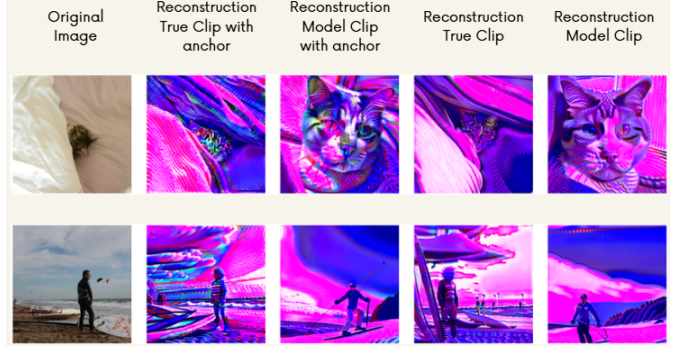


Figure 2: Qualitative Results of Trained Translation Model

5.3 Discussion

The results support the hypothesis that fMRI signals from the higher visual cortex are excellent for semantic decoding but insufficient for perfect spatial reconstruction without additional conditioning.

In summary, the project confirms the validity of the mapping to the Kandinsky-CLIP space. The observed limitations, such as context swapping (e.g., skiing instead of surfing) or inaccuracies in positioning, warrant the future development of a hybrid pipeline that incorporates low-level visual features or region-specific encoders to enhance the spatial coherence of the generated images.

6 Conclusions and Future work

This project demonstrates decoding semantic information from fMRI signals by mapping them to a shared latent space. Our funnel-shaped MLP and Metric Learning strategy outperformed direct regression, successfully capturing the geometry of the Kandinsky-CLIP space and preserving categorical identity.

The analyses conducted confirm that while higher visual areas effectively encode the meaning of objects, semantic information alone is not sufficient for accurate spatial reconstruction. The inaccuracies observed in object localization reflect the "pixel-agnostic" nature of the CLIP mapping, highlighting a trade-off between conceptual coherence and structural fidelity.

Several directions can be explored to improve the proposed framework. A natural extension is evolving the pipeline into a two-way system that combines CLIP semantic embeddings with low-level structural features to guide the diffusion model in spatial arrangement.

Another promising approach is the use of region-specific encoders to integrate signals from the primary visual cortex, which are crucial for recovering edges and spatial coordinates.

Finally, extend the evaluation to other subjects in the NSD dataset and test with larger vision-language models to assess the robustness of cross-modal alignment. [Project](#)

References

- [1] Eric J Allen et al. “A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence”. In: *Nature Neuroscience* (2021). DOI: [10 . 1038 / s41593 - 021 - 00826 - 0](https://doi.org/10.1038/s41593-021-00826-0). URL: [https : / / naturalscenesdataset.org/](https://naturalscenesdataset.org/).
- [2] Aditya Ramesh et al. “Hierarchical Text-Conditional Image Generation with CLIP Latents”. In: *OpenAI* (2022). URL: [https : / / cdn . openai . com / papers / dall-e-2.pdf](https://cdn.openai.com/papers/dall-e-2.pdf).
- [3] Robin Rombach, Andreas Blattmann, et al. “Kandinsky: an Improved Text-to-Image Synthesis Framework”. In: (2023). Technical report.
- [4] P. Scotti. *MindEyeV2: Behavioral Sequences Dataset*. [https : / / huggingface . co / datasets / pscotti / mindeyev2](https://huggingface.co/datasets/pscotti/mindeyev2). 2024.
- [5] Paul S. Scotti et al. “MindEye2: Shared-Task Benchmarks for fMRI-to-Image Reconstruction”. In: *arXiv preprint arXiv:2403.11207* (2024). URL: [https : / / arxiv.org/abs/2403.11207](https://arxiv.org/abs/2403.11207).